

# Taehei Kim

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Google Scholar · Daejeon, South Korea

**Research interests** Mixed reality telepresence; spatial computing; scene synthesis and generation; human-centered XR systems.

## Appointments

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- 2026.09– **Postdoctoral Researcher**, InnoCORE AI-Meta Scientist Program, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea.  
Host: Prof. Tae-Kyun Kim.
- 2024.09– **Co-Founder**, BerryHi, Daejeon, South Korea.  
Mixed reality telepresence systems; shipped *Parameter Room* on the Meta Quest Store.

## Education

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- 2021.03–2026.08 **Ph.D., Culture Technology**, KAIST, Daejeon, South Korea.  
Advisor: Prof. Sung-Hee Lee.  
Dissertation: *Dynamic Space Optimization and Adjoining for Transmitting User and Spatial Context in Mixed Reality Telepresence*.
- 2018.08–2021.02 **M.S., Culture Technology**, KAIST, Daejeon, South Korea.  
Advisor: Prof. Sung-Hee Lee.  
Thesis: *Quadruped Locomotion on Non-Rigid Terrain using Reinforcement Learning*.
- 2013.03–2018.08 **B.A. in Asian Studies** and **B.S. in Computer Science** (dual degree), Yonsei University, Seoul, South Korea.

## Honors and Awards

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- 2026.07 Best Poster Award, Korea Computer Graphics Society (KCGS).
- 2025.11 Excellence Award, Metaverse Developer Contest, Korea Radio Promotion Association (RAPA).
- 2024.10 Best Demo Honorable Mention, IEEE ISMAR.
- 2021.03–2026.08 Fully Funded Ph.D. Fellowship, KAIST.
- 2021.09 Tenacity Scholarship, KAIST.
- 2021.08 First Place, “Hello, World!: New Playground”, Art Center NABI.
- 2018.08–2021.02 Fully Funded M.S. Fellowship, KAIST.

## Publications

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†first author   \*equal contribution

### Peer-Reviewed Journal Articles

- [J4] **Taehei Kim**<sup>†</sup>, J. Shin, H. Kim, H. Jang, J. Kang, and S.-H. Lee. “Voronoi Rooms: Dynamic Visibility Modulation of Overlapping Spaces for Telepresence.” *ACM Transactions on Graphics (TOG)*, 45(2), 2026.  
[doi:10.1145/3777900](https://doi.org/10.1145/3777900). Presented at ACM SIGGRAPH 2026.
- [J3] J. Kang, **Taehei Kim**, H. Kim, and S.-H. Lee. “Real-Time Translation of Upper-Body Gestures to Virtual Avatars in Dissimilar Telepresence Environments.” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 31(10):8711–8725, 2025. [doi:10.1109/TVCG.2025.3577156](https://doi.org/10.1109/TVCG.2025.3577156).
- [J2] **Taehei Kim**<sup>†</sup>, H. Kim, J. Lee, and S.-H. Lee. “Evaluating User Perception toward Physics-Adapted Avatars in Remote Heterogeneous Spaces.” *Computers & Graphics*, 128:104185, 2025. [doi: 10.1016/j.cag.2025.104185](https://doi.org/10.1016/j.cag.2025.104185)
- [J1] D. Yang, J. Kang, **Taehei Kim**, and S.-H. Lee. “Visual Guidance for User Placement in Avatar-Mediated Telepresence between Dissimilar Spaces.” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 30(12):7558–7570, 2024. [doi: 10.1109/TVCG.2024.3354256](https://doi.org/10.1109/TVCG.2024.3354256)

## Peer-Reviewed Full Conference Papers

- [C3] **Taehei Kim**<sup>†</sup>, H. Kim, J. Shin, H. Kim, and S.-H. Lee. “Neighborhood: Dynamic Spatial Alignment for Traversable Telepresence across Remote Spaces.” *ACM Symposium on User Interface Software and Technology (UIST)*, 2026. doi:10.1145/3830398.3830524
- [C2] H. Kim, **Taehei Kim**<sup>\*</sup>, J. Shin<sup>\*</sup>, H. Kim, and S.-H. Lee. “TranSpace: Progressive Anchoring for Metric-Consistent Scene Synthesis.” *ACM International Conference on Multimedia (MM)*, 2026. doi:10.1145/3767308.3836316.
- [C1] J. Kang, D. Yang, **Taehei Kim**, Y. Lee, and S.-H. Lee. “Real-Time Retargeting of Deictic Motion to Virtual Avatars for Augmented Reality Telepresence.” *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 885–893, 2023. doi: 10.1109/ISMAR59233.2023.00104

## Peer-Reviewed Short Papers, Posters, and Demonstrations

- [D4] J. Shin, H. Kim, E. Lee, D. Shin, K. Lee, **Taehei Kim**, H. Kim, J. An, and S.-H. Lee. “Situating Embodied XR Agents via Spatial Reasoning and Prompting.” *IEEE ISMAR Adjunct*, pp. 933–934, 2025. doi:10.1109/ISMAR-Adjunct68609.2025.00255.
- [D3] **Taehei Kim**<sup>†</sup>, J. Shin, H. Kim, H. Jang, J. Kang, and S.-H. Lee. “Visibility Modulation of Aligned Spaces for Multi-User Telepresence.” *IEEE ISMAR Adjunct*, pp. 626–627, 2024. **Best Demo Honorable Mention.**
- [D2] H. Jang<sup>\*</sup>, **Taehei Kim**<sup>\*</sup>, S. Oh, J. Lee, S. Lee, and S. H. Yoon. “Sense of Embodiment Inducement for People with Reduced Lower-Body Mobility and Sensations with Partial-Visuomotor Stimulation.” *ACM SIGGRAPH Emerging Technologies*, pp. 1–2, 2022.
- [D1] H.-I. Kim, **Taehei Kim**, E. Song, S. Y. Oh, D. Kim, and W. Woo. “Multi-Scale Mixed Reality Collaboration for Digital Twin.” *IEEE ISMAR Adjunct*, pp. 435–436, 2021.

## Preprints and Manuscripts under Review

- [P2] One paper currently under review, 2026.
- [P1] **Taehei Kim**<sup>†</sup> and S.-H. Lee. “Quadruped Locomotion on Non-Rigid Terrain using Reinforcement Learning.” *arXiv:2107.02955*, 2021.

## Grants and Funded Research Projects

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|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 2026.06–        | KEP Startup Fund — <i>Real-Space Overlay Scene Generation for VR Crowdsourcing Data Collection</i> . USD 37,000. Role: Internal director. |
| 2024.05–2025.12 | Korea Radio Promotion Association (RAPA) — <i>Mixed Reality Telepresence System</i> . Role: Student project lead).                        |
| 2024.08–2024.09 | Daejeon Art Center Joint New Technology Performance. USD 11,000. Role: Lead director.                                                     |
| 2022.03–2024.02 | National Research Foundation of Korea — <i>Mixed Reality Scenes in Heterogeneous Spaces</i> . Role: Student project lead.                 |
| 2021.03–2022.02 | Korea Institute for Advancement of Technology (KIAT) — VR/AR Expert Training Program. Role: Participating researcher.                     |

## Public Demonstrations, Exhibitions, and Deployed Systems

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| 2025.11 | <i>Parameter Room</i> , Mixed reality telepresence application, Meta Quest Store (BerryHi).                                                             |
| 2024.10 | <i>Visibility Modulation of Aligned Spaces for Multi-User Telepresence</i> , IEEE ISMAR 2024 Demonstrations, Seattle, USA. Best Demo Honorable Mention. |
| 2024.09 | X-Space, Daejeon Art Center, Daejeon, South Korea.                                                                                                      |
| 2022.07 | <i>Sense of Embodiment Inducement with Partial-Visuomotor Stimulation</i> , ACM SIGGRAPH 2022 Emerging Technologies, Vancouver, Canada.                 |

## Teaching Experience

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### Guest Lectures and Instruction

- 2021.08–2023.07 Metaverse Instructor. Designed and delivered XR content-creation curricula for public and corporate audiences. Clients: Dangjin Cultural Foundation, Daejeon Culture and Arts Foundation, Hanwha Galleria, 3PROTV.

## Courses Prepared to Teach

Computer Graphics; Virtual and Augmented Reality; Human–Computer Interaction; Generative Models and Agents for 3D Content Generation.

## Mentoring

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2024.03– Mentored M.S. students (Hyeslim Kim, Jihun Shin, Hyeonjin Kim) at KAIST LAVA Lab and in collaborating laboratories (Taeyeon Kim). Mentorship resulted in co-authored publications [J4], [J2], [C3], [C2], [D4], [D3].

## Technical Skills

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*Programming* C#, Python, C++.

*XR development* Unity, Mixed Reality Toolkit (MRTK), Photon networking, Meta Quest 3, HTC Vive.

*Methods* Computational geometry and optimization, LLM/diffusion-based scene synthesis, quantitative and qualitative HCI evaluation.

## Languages

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Korean (native); English (fluent).

## References

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**Sung-Hee Lee**, Professor, Graduate School of Culture Technology, KAIST, [sunghee.lee@kaist.ac.kr](mailto:sunghee.lee@kaist.ac.kr) (Ph.D. advisor).

**Tae-Kyun Kim**, Professor, School of Computing, KAIST, [kimtaekyun@kaist.ac.kr](mailto:kimtaekyun@kaist.ac.kr) (Postdoctoral host).